T1,

Q1, $x: 2.3m$ $y: 1.7m$ $z: 7.4m$ $\phi_a: 32^\circ$ $\phi_o: 178^\circ$ $\phi_n: 4^\circ$

The transformation matrix for the end effector is:

TODO:
推导一遍

$$
{}^{b}T_{H} = \begin{bmatrix} C\phi_a C\phi_o & C\phi_a S\phi_o S\phi_n - S\phi_a C\phi_n & C\phi_a S\phi_o C\phi_n + S\phi_a S\phi_n & P_x \\ S\phi_a C\phi_o & S\phi_a S\phi_o S\phi_n + C\phi_a C\phi_n & S\phi_a S\phi_o C\phi_n - C\phi_a S\phi_n & P_y \\ -S\phi_o & C\phi_o S\phi_n & C\phi_o C\phi_n & P_z \\ 0 & 0 & 0 & 1 \end{bmatrix}
$$

$$
= \begin{bmatrix} -0.848 & -0.527 & 0.066 & 2.3 \\ -0.530 & 0.847 & -0.041 & 1.7 \\ -0.035 & -0.070 & -0.997 & 7.4 \\ 0 & 0 & 0 & 1 \end{bmatrix}
$$

Therefore, the orientation is:

$$
\underline{noa} = \begin{bmatrix} -0.848 & -0.527 & 0.066 \\ -0.530 & 0.847 & -0.041 \\ -0.035 & -0.070 & -0.997 \end{bmatrix}
$$

The position is:

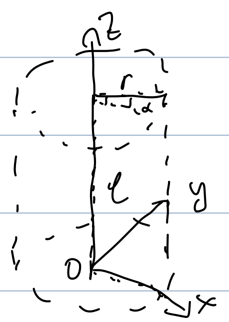
$$
\underline{P} = \begin{bmatrix} 2.3 \\ 1.7 \\ 7.4 \end{bmatrix}
$$

Q2. Given ${}^{U}T_{H} = \begin{bmatrix} 0.460 & -0.861 & 0.220 & 2.441 \\ 0.859 & 0.494 & 0.137 & 4.591 \\ -0.226 & 0.126 & 0.966 & 8.200 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

the position vector is: $\underline{p} = \begin{bmatrix} 2.441 \\ 4.591 \\ 8.200 \end{bmatrix}$

the orientation vector is: $\underline{noa} = \begin{bmatrix} 0.460 & -0.861 & 0.220 \\ 0.859 & 0.494 & 0.137 \\ -0.226 & 0.126 & 0.966 \end{bmatrix}$

Therefore:



$\underline{p} = \begin{bmatrix} rC\alpha \\ rS\alpha \\ l \end{bmatrix} \Rightarrow \begin{cases} r\cos\alpha = 2.441 \\ r\sin\alpha = 4.591 \\ l = 8.200 \end{cases}$

$\Rightarrow \begin{cases} l = 8.200 \quad m \\ r = 5.200 \quad m \\ \alpha = 62.00^\circ \end{cases}$

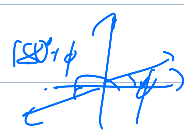
From $Rot^{-1}(a,\phi)\,\underline{noa} = Rot(o,\theta)\,Rot(a,\psi)$

$\Rightarrow \begin{bmatrix} C\phi & S\phi & & \\ -S\phi & C\phi & & \\ & & 1 & \\ & & & 1 \end{bmatrix} \begin{bmatrix} n_x & o_x & a_x & \\ n_y & o_y & a_y & \\ n_z & o_z & a_z & \\ & & & 1 \end{bmatrix}$

$$= \begin{bmatrix} C_\theta & & S_\theta & \\ & 1 & & \\ -S_\theta & & C_\theta & \\ & & & 1 \end{bmatrix} \begin{bmatrix} C\psi & -S\psi & & \\ S\psi & C\psi & & \\ & & 1 & \\ & & & 1 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} n_x C_\phi + n_y S_\phi & o_x C_\phi + o_y S_\phi & a_x C_\phi + a_y S_\phi & 0 \\ -n_x S_\phi + n_y C_\phi & -o_x S_\phi + o_y C_\phi & -a_x S_\phi + a_y C_\phi & 0 \\ n_z & o_z & a_z & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} C_\theta C_\psi & -C_\theta S_\psi & S_\theta & 0 \\ S_\psi & C_\psi & 0 & 0 \\ -S_\theta C_\psi & S_\theta S_\psi & C_\theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



$$\Rightarrow \tan\phi = \frac{a_y}{a_x} \Rightarrow \phi_1 = \tan^{-1}\left(\frac{a_y}{a_x}\right) = 31.91^\circ$$

$$\text{or } \phi_2 = \tan^{-1}\left(\frac{-a_y}{-a_x}\right) = 211.9^\circ$$

$$\tan\theta = \frac{a_x C_\phi + a_y S_\phi}{a_z} \Rightarrow \theta_1 = 15.0^\circ$$

$$\text{or } \theta_2 = -15.0^\circ$$

$$\tan\psi = \frac{-n_x S_\phi + n_y C_\phi}{-o_x S_\phi + o_y C_\phi} \Rightarrow \psi_1 = 29.1^\circ$$

$$\text{or } \psi_2 = 209.1^\circ$$

Q3. (a) Cylindrical Manipulator:

$$T_{cyl}(r, \alpha, l) =$$

$$\begin{bmatrix} 1 & & & \\ & 1 & & \\ & & 1 & l \\ & & & 1 \end{bmatrix} \cdot \begin{bmatrix} C\alpha & -S\alpha & & \\ S\alpha & C\alpha & & \\ & & 1 & \\ & & & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & & & r \\ & 1 & & \\ & & 1 & \\ & & & 1 \end{bmatrix}$$

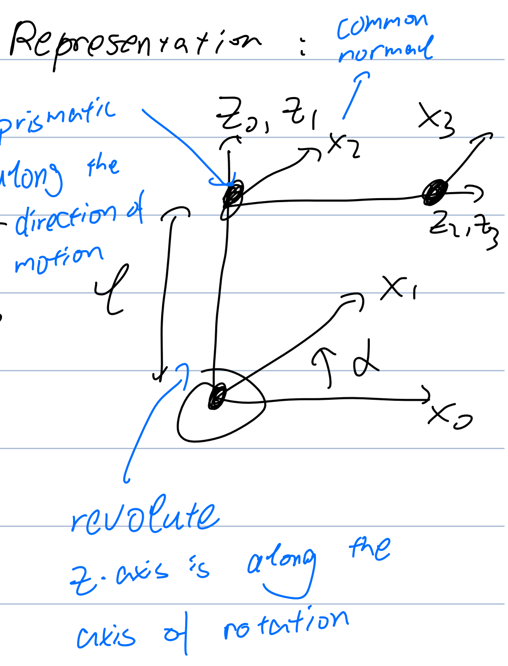
$$= \begin{bmatrix} C\alpha & -S\alpha & & \\ S\alpha & C\alpha & & \\ & & 1 & \\ & & & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & & & r \\ & 1 & & 0 \\ & & 1 & l \\ & & & 1 \end{bmatrix}$$

$$= \begin{bmatrix} C\alpha & -S\alpha & 0 & rC\alpha \\ S\alpha & C\alpha & 0 & rS\alpha \\ 0 & 0 & 1 & l \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

(b) The Denavit-Hartenberg Representation:

D-H table:

#	θ	d	a	α
1	α	0	0	0
2	0	l	0	90°
3	0	r	0	0



$$A_1 = Rot(a, \alpha) = \begin{bmatrix} C\alpha & -S\alpha & & \\ S\alpha & C\alpha & & \\ & & 1 & \\ & & & 1 \end{bmatrix}$$

$$A_2 : Rot(n, 90^\circ) \cdot Trans(0, 0, \ell)$$

$$= \begin{bmatrix} 1 & & & \\ & C & -S & \\ & S & C & \\ & & & 1 \end{bmatrix} \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & 1 & \ell \\ & & & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & & & \\ & 0 & -1 & \\ & 1 & 0 & \ell \\ & & & 1 \end{bmatrix}$$

$$A_3 = \begin{bmatrix} 1 & & & 0 \\ & 1 & & 0 \\ & & 1 & r \\ & & & 1 \end{bmatrix}$$

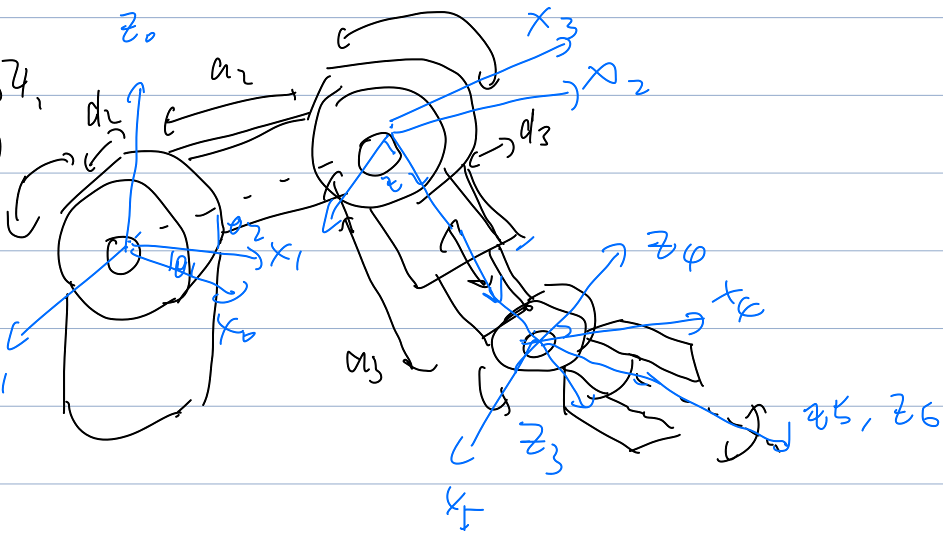
$$^{U}T_H = A_1 A_2 A_3 = \begin{bmatrix} C\alpha & -S\alpha & & \\ S\alpha & C\alpha & & \\ & & 1 & \\ & & & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & & & \\ & 0 & -1 & \\ & 1 & & \ell \\ & & & 1 \end{bmatrix}$$

$$\cdot \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & 1 & r \\ & & & 1 \end{bmatrix}$$

$$= \begin{bmatrix} C\alpha & 0 & S\alpha & 0 \\ S\alpha & 0 & -C\alpha & 0 \\ 0 & 1 & 0 & l \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & 1 & r \\ & & & 1 \end{bmatrix}$$

$$= \begin{bmatrix} C\alpha & 0 & S\alpha & rS\alpha \\ S\alpha & 0 & -C\alpha & -rC\alpha \\ 0 & 1 & 0 & l \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

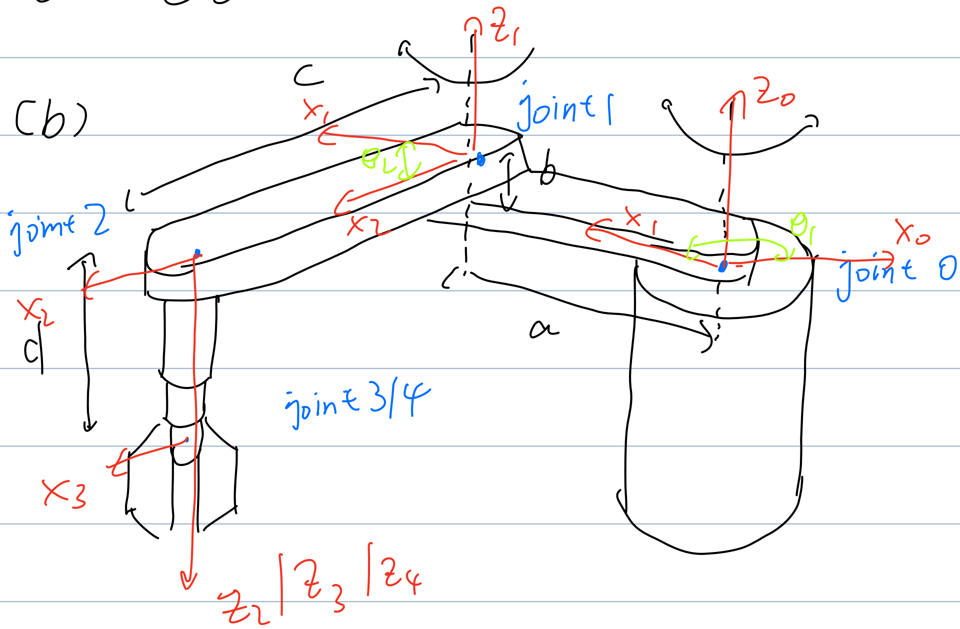
Q4, (a)



#	θ	d	a	α
1	θ_1	0	0	90°
2	θ_2	$-d_2$	a_2	0°
3	θ_3	d_3	0	90°
4	0°	a_3	0	90°

5	θ_5	0	0	90°
6	θ_6	0	0	0

(b)



#	θ	d	a	α
1	θ_1	b	a	0
2	θ_2	0	c	180°
3	0	d	0	0
4	θ_4	0	0	0

$$A_1 = \mathrm{Rot}(a, \theta_1) \cdot \mathrm{Trans}(a, 0, b)$$

$$= \begin{bmatrix} c\theta_1 & -s\theta_1 & & \\ s\theta_1 & c\theta_1 & & \\ & & 1 & \\ & & & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & & & a \\ & 1 & & \\ & & 1 & b \\ & & & 1 \end{bmatrix} = \begin{bmatrix} c\theta_1 & -s\theta_1 & & a c\theta_1 \\ s\theta_1 & c\theta_1 & & a s\theta_1 \\ & & 1 & b \\ & & & 1 \end{bmatrix}$$

$$A_2 = Rot(a, \theta_2)\cdot Trans(c, 0, 0)\cdot Rot(n, 180^\circ)$$

$$= \begin{bmatrix} C\theta_2 & -S\theta_2 & & \\ S\theta_2 & C\theta_2 & & \\ & & 1 & \\ & & & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & & & c \\ & 1 & & \\ & & 1 & \\ & & & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & & & \cdot \\ & -1 & 0 & \\ & 0 & -1 & \\ & & & 1 \end{bmatrix}$$

$$= \begin{bmatrix} C\theta_2 & -S\theta_2 & & \\ S\theta_2 & C\theta_2 & & \\ & & 1 & \\ & & & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & & & c \\ & -1 & & \\ & & -1 & \\ & & & 1 \end{bmatrix}$$

$$= \begin{bmatrix} C\theta_2 & S\theta_2 & & cC\theta_2 \\ S\theta_2 & -C\theta_2 & & cS\theta_2 \\ & & -1 & \\ & & & 1 \end{bmatrix}$$

$$A_3 = Trans(0, 0, d) = \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & 1 & d \\ & & & 1 \end{bmatrix}$$

$$A_4 = Rot(a, \theta_4) = \begin{bmatrix} C\theta_4 & -S\theta_4 & & \\ S\theta_4 & C\theta_4 & & \\ & & 1 & \\ & & & 1 \end{bmatrix}$$

$${}^{U}T_{H} = A_1 \cdot A_2 \cdot A_3 \cdot A_4.$$

$$= \begin{bmatrix} C\theta_1 & -S\theta_1 & & aC\theta_1 \\ S\theta_1 & C\theta_1 & & aS\theta_1 \\ & & 1 & b \\ & & & 1 \end{bmatrix} \begin{bmatrix} C\theta_2 & S\theta_2 & & cC\theta_2 \\ S\theta_2 & -C\theta_2 & & cS\theta_2 \\ & & -1 & \\ & & & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & & & \\ & 1 & & \\ & & 1 & d \\ & & & 1 \end{bmatrix} \begin{bmatrix} C\theta_4 & -S\theta_4 & & \\ S\theta_4 & C\theta_4 & & \\ & & 1 & \\ & & & 1 \end{bmatrix}$$

$$= \begin{bmatrix} C\theta_1 C\theta_2 - S\theta_1 S\theta_2 & C\theta_1 S\theta_2 + S\theta_1 C\theta_2 & 0 & cC\theta_1 C\theta_2 - cS\theta_1 S\theta_2 + aC\theta_1 \\ S\theta_1 C\theta_2 + C\theta_1 S\theta_2 & S\theta_1 S\theta_2 - C\theta_1 C\theta_2 & 0 & cS\theta_1 C\theta_2 + cC\theta_1 S\theta_2 + aS\theta_1 \\ & & -1 & b \\ & & & 1 \end{bmatrix}$$

$$\begin{bmatrix} A_3 \end{bmatrix} \begin{bmatrix} A_4 \end{bmatrix}$$

$$= \begin{bmatrix} C_{12} & S_{12} & 0 & cC_{12} + aC_1 \\ S_{12} & -C_{12} & & cS_{12} + aS_1 \\ & & -1 & b-d \\ & & & 1 \end{bmatrix}$$

$$\begin{bmatrix} C\theta_4 & -S\theta_4 & & \\ S\theta_4 & C\theta_4 & & \\ & & 1 & \\ & & & 1 \end{bmatrix}$$

$$= \begin{bmatrix} C_{1+2}C_4 + S_{1+2}S_4 & -C_{1+2}S_4 + S_{1+2}C_4 & 0 & cC_{1+2} + aC_1 \\ S_{1+2}C_4 - C_{1+2}S_4 & -S_{1+2}S_4 - C_{1+2}C_4 & 0 & aS_{1+2} + aS_1 \\ 0 & 0 & -1 & b-d \\ & & & 1 \end{bmatrix}$$

$$\begin{bmatrix} C_{1+2-4} & S_{1+2-4} & 0 & cC_{1+2} + aC_1 \\ S_{1+2-4} & -C_{1+2-4} & 0 & aS_{1+2} + aS_1 \\ & & -1 & b-d \\ & & & 1 \end{bmatrix}$$